

# Zihao Wang

404-952-0911 | [ziw140@ucsd.edu](mailto:ziw140@ucsd.edu) | [gdfwj.github.io](https://gdfwj.github.io) | [github.com/gdfwj](https://github.com/gdfwj)

## EDUCATION

---

### University of California, San Diego

*Master of Science in Computer Science*

La Jolla, CA

Sep. 2024 – Now

- GPA: 4.0/4.0
- Key courses: Systems for LLMs and AI Agents, Differentiable Programming (A+), ML: Learning Algorithms (A+)

### Beihang University

Beijing, China

*Bachelor of Engineering in Computer Science (Minor in Mathematics and Applied Mathematics)* Sep. 2020 – Jun. 2024

- GPA: 3.92/4.0
- Key courses: Computer Organization (96), Operating Systems (95), Object Oriented Design Construction (98), Compiler Technology (95), Signal and System (98), Computer Network Experiments (97)

## RESEARCH INTERESTS

---

My research interests lie in Machine Learning, Systems for AI Agents, IoT (Internet of Things), and edge computing. Specifically, I focus on building context-aware multimodal reasoning systems that integrate LLMs with sensor-based data for efficient on-device intelligence. I am currently working on AI for IoT and efficiency AI Agent systems on edge devices in SeeLab, advised by Prof. Tajana Šimunić Rosing and Ye Tian.

## SCHOLARSHIPS & AWARDS

---

- **2021, 2022, 2023** Academic Merit Scholarship (Awarded to undergraduate students for excelling GPA throughout the school year)
- **2022, 2023** Discipline Competition Scholarship (Awarded to undergraduate students who achieved outstanding rankings in national or university-level discipline competitions)

## PUBLICATIONS

---

1. Ye Tian\*, **Zihao Wang\***, Onat Gungor, Xiaoran Fan, Tajana Rosing. “MultiLifeQA: A Multi-Domain Health Reasoning Benchmark.” *ICLR*, 2026 (under review)
2. Ye Tian, Xiaoyuan Ren, **Zihao Wang**, Onat Gungor, Xiaofan Yu, Tajana Rosing. “DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs.” *22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025

## RESEARCH EXPERIENCE

---

### Predictive Tool-Call & Co-Scheduling for Edge LLM Agents

Oct, 2025–Present

*Research Assistant, Advisors: Ye Tian and Tajana Rosing*

System Energy Efficiency Lab, University of California, San Diego

- Proposed an agent-oriented runtime that decouples *CPU-side tool calls/DB queries* from *GPU-side LLM prefill/decoding* to reduce end-to-end latency on edge devices.
- Extended the **CHA** framework (<https://github.com/Institute4FutureHealth/CHA>) to support multi-step tool–LLM loops and basic timestamped logging; early prototype runs succeed end-to-end.
- Designed **predictive tool-call speculation** (branch-prediction style) for read-only/idempotent tools with confidence gating and budget control; initial prototype under development.
- Sketched a lightweight co-scheduling runtime (asyncio tools + GPU inference loop) and prepared reproducible workloads via SQL-based “sensor” queries.
- *In progress*: system write-up and open-source repo plan (target venues: MobiCom / DAC).

## Multidimensional Health QA Benchmark (MultiLifeQA)

Jul, 2025–Sep, 2025

*Research Assistant, Advisors: Ye Tian and Tajana Rosing*

System Energy Efficiency Lab, University of California, San Diego

- Proposed a large-scale QA benchmark comprising 22,573 questions for multidimensional health reasoning using database-stored lifestyle data.
- Designed a framework that transforms structured datasets (e.g., Excel files) into a MySQL database and automatically generates QA pairs from question templates and database entries.
- Developed context-based and database-augmented evaluation protocols with fine-grained metrics for validity, execution quality, and answer accuracy.
- Led experiments on eight open-source and three proprietary LLMs to analyze their reasoning capabilities and limitations in multidimensional health understanding, and released the full dataset and benchmark at <https://github.com/gdfwj/MultilifeQA>.
- Paper submitted to **ICLR 2026** (under review).

## DailyLLM: Multimodal Activity Log Generation

Jan, 2025–April, 2025

*Research Assistant, Advisors: Ye Tian and Tajana Rosing*

System Energy Efficiency Lab, University of California, San Diego

- Proposed **DailyLLM**, a lightweight LLM-based system for context-aware activity log generation and summarization using multimodal smartphone and smartwatch sensors.
- Designed and generated the IMU-based dataset for activity recognition, including data preprocessing, labeling, and augmentation.
- Implemented fine-tuning and evaluation pipelines for all LLM components and conducted controlled comparisons against baselines.
- Led the full experimental design and execution, and developed the **project website** for dataset and code release.
- Paper accepted to **The 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2025)**.

## Reconstruction of perceived face images from fMRI

Apr, 2023–Dec, 2023

*Research Assistant, Advisors: Hui Zhang*

Medical-Engineering Innovation Research Institute

- Designed and trained a model to reconstruct the face image perceived in brain from fMRI data.
- Implemented a pre-training strategy on the CelebA dataset, utilizing the features extracted by VGGFace as conditions, as a solution to the scarcity of fMRI-face paired data
- Designed a double-flow architecture of the discriminator for GAN to challenge the generator.
- Manuscript was published on arXiv.

## TECHNICAL SKILLS

---

**Languages:** Java, Python, C/C++, SQL (MySQL), Verilog

**Frameworks:** PyTorch, gradio, OpenGL

**Developer Tools:** Git, Docker, VS Code, Kubernetes